

Education

Swarthmore College, Philadelphia, PA Aug 2023—May 2027
Double-Major: BS in General Engineering (electrical engineering focus), BA in Applied Mathematics
Relevant coursework:

- Electronic circuit analysis & applications, Digital signal processing, Data structures & algorithms (C++)
- Ordinary & Partial differential equations (PDEs), Numerical PDEs, Real Analysis, Tensor decompositions
- Thermo-fluid mechanics, Electromagnetism, waves and optics with biomedical applications.

Teaching Assistant for Electrical Circuit Analysis (ENGR 011) Aug 2025—present
Swarthmore Varsity Men’s Soccer Aug 2023—present
Overall GPA: 3.98/4.00

Research and work experience

Advanced Materials Intern & Research Assistant – US Naval Research Laboratory, Washington, DC May 2025-Aug 2025

- Measured thermal diffusivity and specific heat of thermoelectric semiconductors using laser flash analysis (LFA).
- Built Arduino-controlled LN₂ dispenser with closed-loop thermocouple monitoring, automating LFA experiments
- Created graphical-user-interface and data-logging interfacing between LN₂ dispenser and lab computer.
- Developed experiments to improve the lab’s measurement procedure for IR-transparent and high-resistance samples.
- Poster presentations: Swarthmore *Sigma Xi* 2025 poster session.

Applied Mathematics Research Assistant – Swarthmore College Mathematics, Philadelphia, PA Jan 2024-present

- Utilizing computational fluid dynamics and numerical methods to research high-order accurate methods for partial differential equations (PDEs), plasma/kinetic models in MATLAB.
- Developing low-rank, structure-preserving integrator for Vlasov-Fokker-Planck plasma equation.
- Poster presentations: 2025 SIAM NNP conference, 2024 & 2025 *Sigma Xi* Swarthmore poster sessions.

Electrical Engineering Research Assistant – Swarthmore College Engineering, Philadelphia, PA Nov 2023—May 2024

- Researched electrical/aerospace technology behind wind-energy devices to develop oscillatory wind-energy harvester.
- Used Arduino, MATLAB & ViscousFlow to simulate air vortex-shedding & electrical induction power output.

Software Engineering Summer Intern - Oregon Health and Science University, Portland, OR Jun 2022—Aug 2022

- Used Kotlin to develop Android app that detects heart murmurs using audio data from a Bluetooth stethoscope.
- Presented machine-learning paper to the lab’s reading group.

Projects

Analog Circuit and PCB Design: used Fusion360 to design and fabricate printed circuit boards (PCBs) Aug 2025—Dec 2025

- Combined MOSFETs, sensors, actuators, op-amps, comparators, audio amplifiers, and other integrated circuits
- Projects: [audio equalizer](#), [temp-controlled fan](#), [light-optimizing solar panel](#), [regulated DC power supply](#).

AI Python Stock Trading: trained neural network on technical indicators to trade stocks using Alpaca API, [link](#) Feb 2025

Generative Adversarial Network (GAN): Image-generating model trained on web-scraped art, [link](#) Apr 2022—Feb 2023

FireSale: Used Java, AWS backend to build Android app to reduce food waste and hunger, [link](#) 2020—2021

Technical & Professional Skills

Technical: SPICE, PCB design, CAD, Fusion360, Soldering, Embedded systems, PDE solvers, Numerical methods
Programming: Python, MATLAB, Java, C++, HTML/JavaScript, LaTeX
Software: VSCode, MATLAB, Arduino, Git, SolidWorks, AutoCAD, NumPy, TensorFlow
Languages: Spanish (Fluent), Global Seal of Biliiteracy (2022), **Citizenship:** USA, Ireland

Honors and Scholarships

Tau Beta Pi Engineering Honors Society Mar 2026
Delaware Valley Engineers Undergraduate Scholarship, Delaware Valley Engineers Society Feb 2025
Donna Prentice Memorial Scholarship, American Society of Civil Engineers Feb 2024
National Merit Scholarship, National Merit Scholarship Corporation Apr 2023